**Higher National Diploma inEngineering (Civil)**
**4th Year, First Semester Examination – 2020**
**CE3223 – Water Supply & Waste Treatment**

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| Instructions for Candidates: | No. ofquestions :06 |
| **Answer five questions only** | No. of pages : 04 |
| **All questions carry equal marks** | Time :3 hrs |

Q1).

a) What are the factors which can be considered as determinants of water demand? (4 marks)

b) What doyou mean by per capita demand in water? (2 marks)

c) Population forecasting is widely use in water supply designs. Among many methods of population forecasting, two significant methods are "Arithmetical increase method" and Geometrical increase method". The population data of a town from census records are given as follows. Find the population after three decades by using Arithmetical increase method.

| Year | Population | Increase in population |
|---|---|---|
| 1950 | 100000 | - |
| 1960 | 109000 | 9000 |
| 1970 | 116600 | 7600 |
| 1980 | 128200 | 11600 |

(4 marks)

d) Use the same data given above to calculate the population after 3 decades by using Geometrical increase method. State your assumptions. (5 marks)

e) Name five surface water sources. (5 marks)

(Total 20 marks)

Q2).

a) What are the main types of geological formations of ground water? Give one example for each type. (4 marks)

b) *Capacity of a reservoir can be defined as the gross volume of water which can be* stored in it. By using contour lines, reservoir capacity can be determined. From a contour plan of a proposed reservoir, the following data were found. Calculate the volume of the reservoir.

| **Contour (m)** | 200 | 210 | 220 | 230 | 240 |
|---|---|---|---|---|---|
| **Area of contour ($m^2$)** | 3 | 5 | 10 | 21 | 30 |

(4 marks)

c) Name four types of microorganisms that can be found in nature. (4 marks)

d) Define the term "Eutrophication" and mention three contributing factors for causing eutrophication. (4 marks)

e) Define the following terms use in water supply engineering.

i. Infiltration

ii. Percolation

iii. Transpiration

iv. Evaporation 4 marks)

(Total 20 marks)

3).

a) Draw a flow chart highlighting the main processes in water treatment process. (4 marks)

b) Write the procedure of conducting "Jar Test" mentioning its purpose. (4 marks)

c) Name four troubles in sand filters. (4 marks)

d) Name four factors to be considered when designing water transmission lines. (4 marks)

e) Name and draw two types of layouts in water distribution systems. (4 marks)

(Total 20 marks)

Q4).

a) Briefly describe about the four types of joints in sewer. (6 marks)

b) Compare and contrast "Trickling filters" and "Activated sludge system" in wastewater treatment. (4 marks)

c) Calculate 5-day BOD value of the sample consists following data.

Sewage volume= 20 ml, water volume = 200 ml,

DO at the beginning = 10 mg/l, final DO = 2 mg/l (3 marks)

d) Draw the oxygen sag curve and mark important points on it. (3 marks)

e) Define the following terms use in waste management.

i. Sewage

ii. Refuse

iii. Garbage

iv. Sewer (4 marks)

(Total 20 marks)

5).

a) List four types of sewer sections. (4 marks)

b) A rectangular sedimentation tank for a city water treatment system need to be designed by considering design details given by the relevant authority. The city must treat maximum flow rate of $0.5 m^3/s$ and the design over flow rate of $32.5\ m^3/d.m^2$. The tank has the dimensions of length: width of6:1 and the depth of 3000 mm.

i. Calculate the dimensions of the tank.

ii. Calculate the detention time of the tank. (6 marks)

c) Chlorine can be used as a disinfectant in water treatment. Write two advantages and disadvantages of chlorine as a disinfectant. (4 marks)

d) Explain F/M ratio with its use. (4 marks)

e) Write two requirements of a domestic septic tank. (2 marks)

(Total 20 marks)

Q6).

a) Siphon spillway is a type of a storm water regulating. Describe about siphon spillway by using suitable sketches. (5 marks)

b) List and briefly describe about functional elements of waste disposal. (5 marks)

c) Write five adverse health effects cause due to noise pollution. (5 marks)

d) ISO certification is a certificate issued to companies, industries…, etc. internationally. What is the reason for issuing ISO 9001 certification for a company? Write four benefits of having ISO 9001 certification for a company representing building construction. (5 marks)

(Total 20 marks)